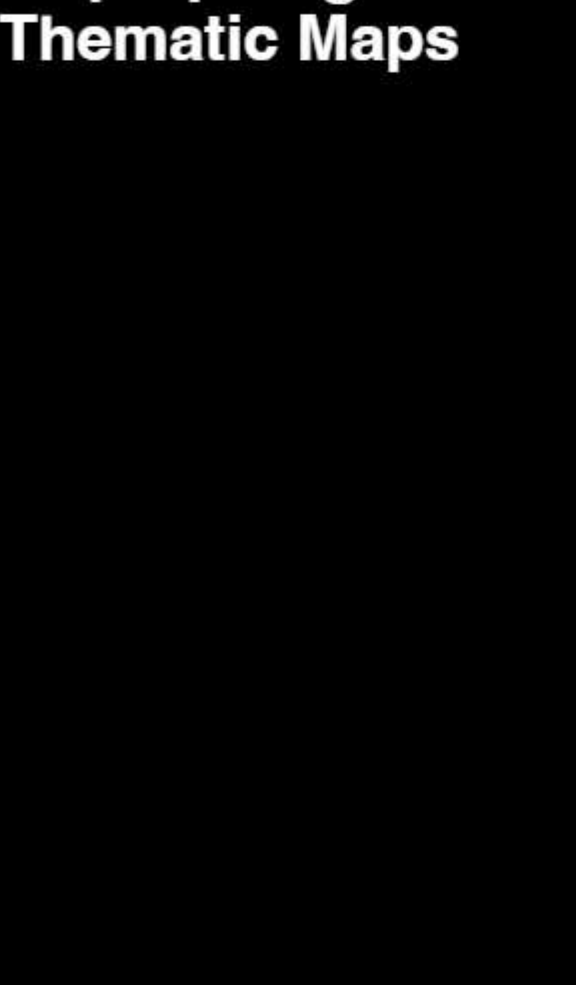


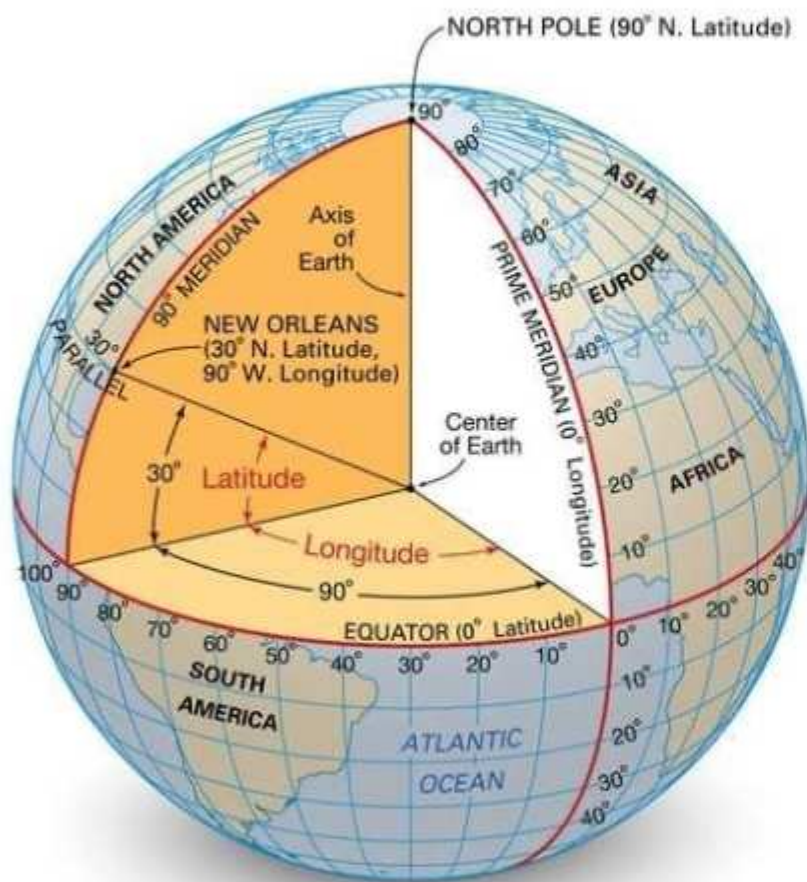
Map Spotlight 4: Thematic Maps



New and Correct Chart Showing the Variations
of the Compass, 1702, Edmond Halley



1. Authalic Sphere (標準球體)



© Encyclopædia Britannica, Inc.

- Unit of latitude and longitude
- DMS format (degree, min, sec)
 - 45° 33' 37" N, 94° 9' 44" W
- Decimal degree
 - -94.162222, 45.560278
- Transformation

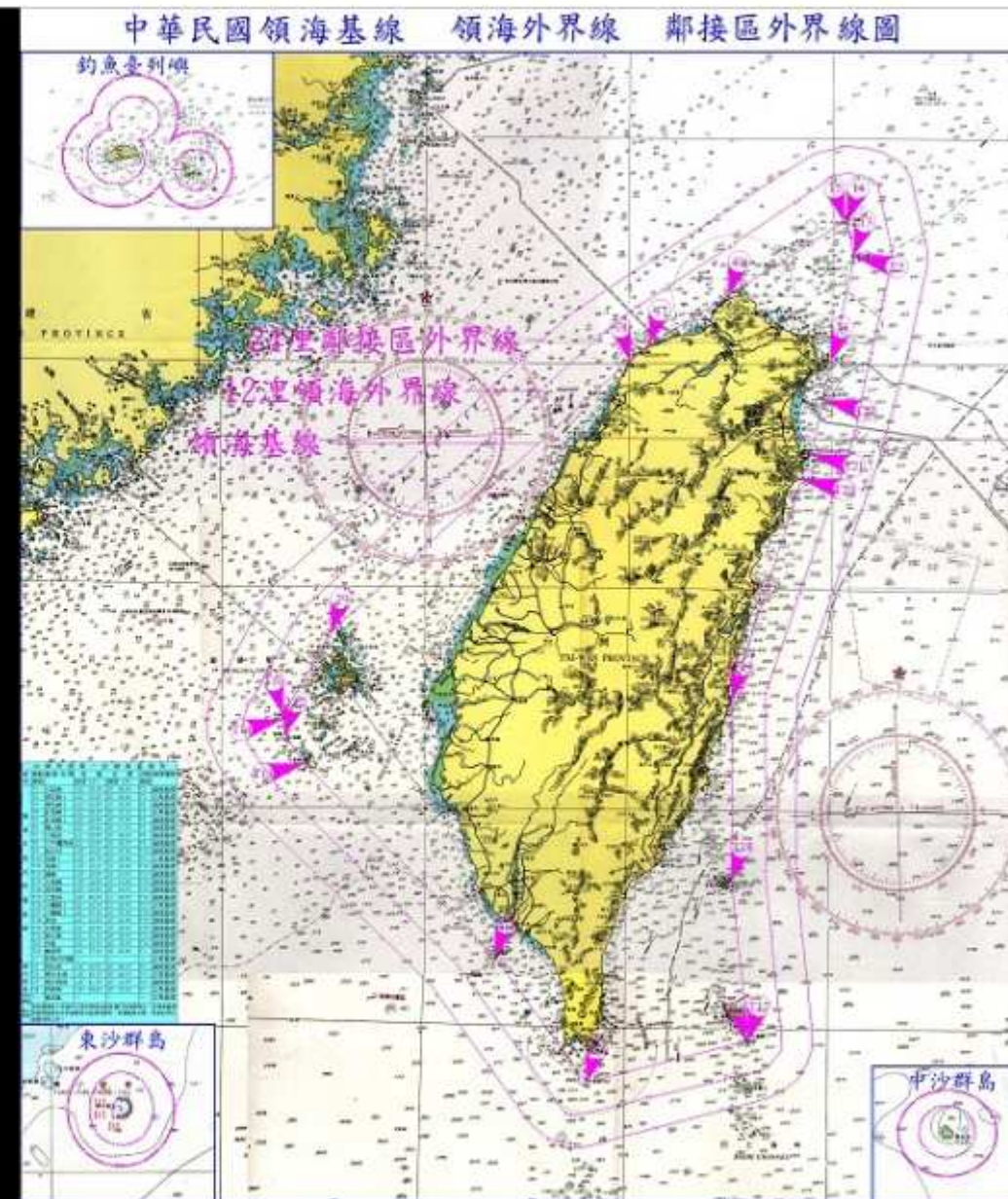
若有一經緯度是：a度、b分、c秒，以十進位表示及的公式如下：

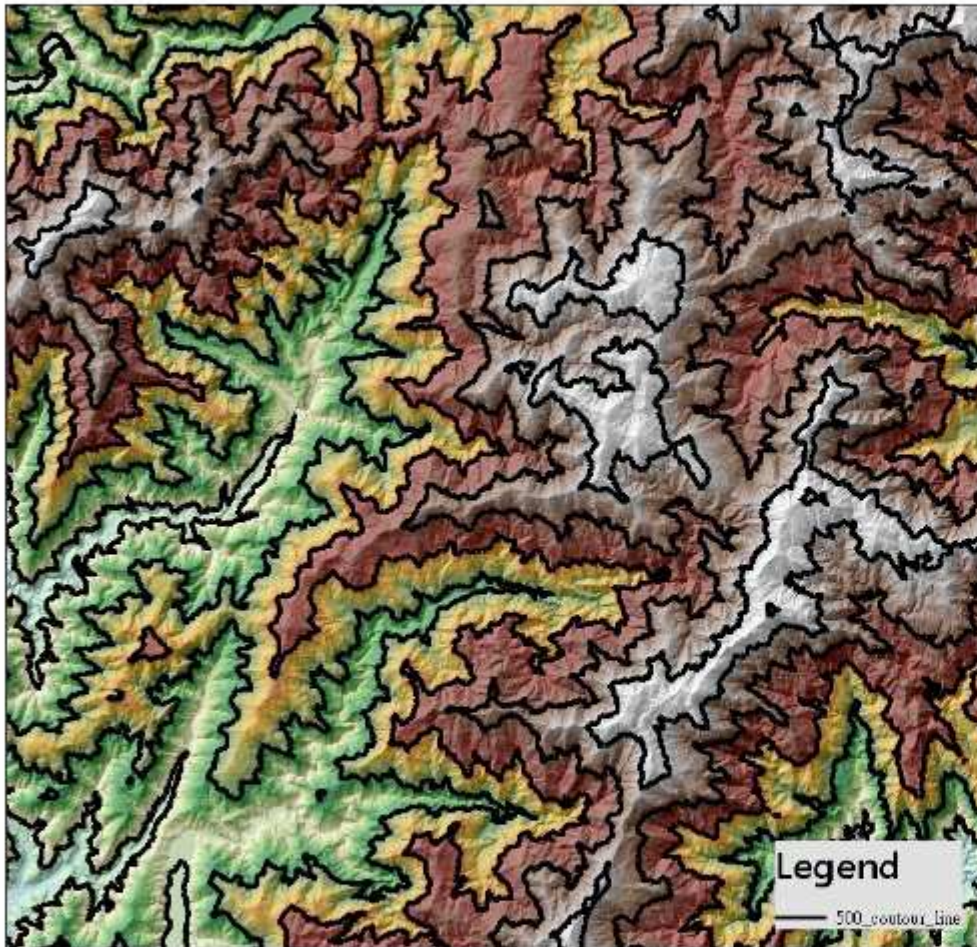
$$\text{十進制} = a + \frac{b}{60} + \frac{c}{3600}$$

$$\begin{aligned} & 121^{\circ} 18' 32'' \text{ E} \\ &= 121 + 18/60 + 32/3600 \\ &= 121.3089 \end{aligned}$$

An arc minute for latitude:
 $40008 \text{ km} \div 360 (\text{degree}) \div 60 (\text{minute}) \approx$
 1.852 km/minute
 $= 1 \text{ Nautical mile (NM) (浬)}$

An arc second for longitude:
 $36752 \text{ km} \div 360 (\text{degree}) \div 60 (\text{minute}) \div$
 $60 (\text{second}) \approx$
 0.0283 km/second





- **Title**
- **Compass**
- **Scale bar unit**
- **Grid**
- **DEM legend**
- **Contour line label**
- **Metadata**

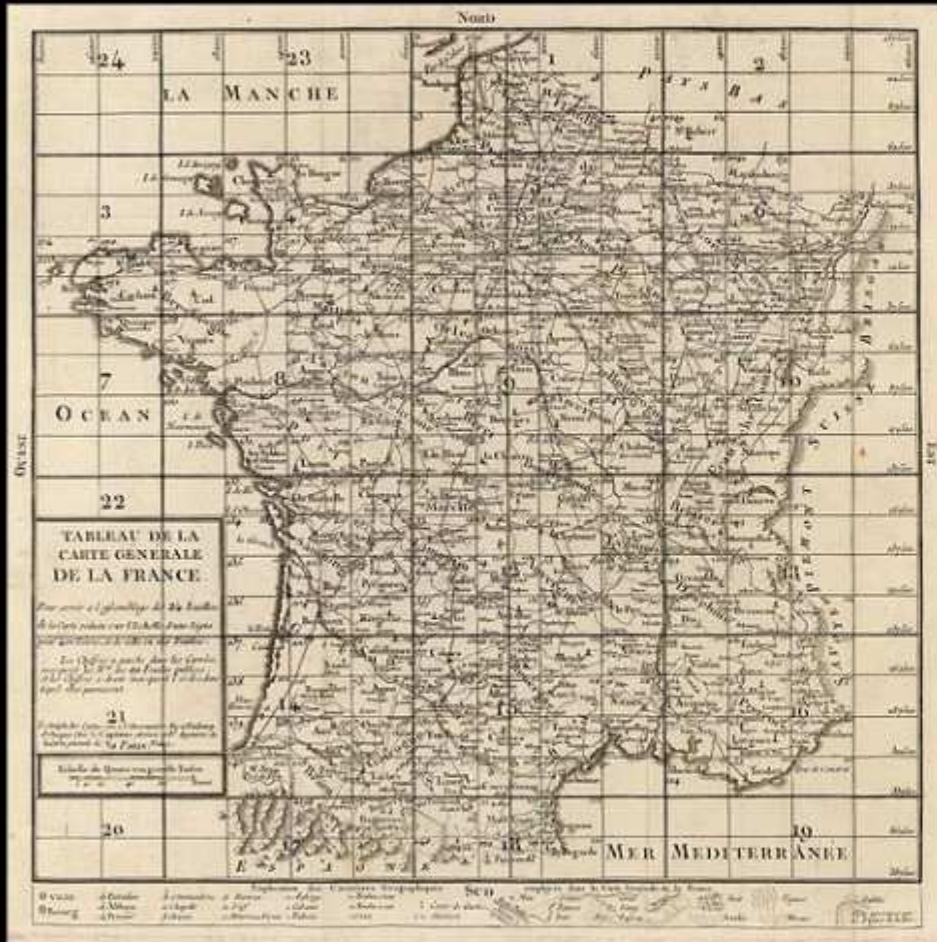
This Week

- The nature of geographic data
 - Spatial autocorrelation (空間自相關), distance decay (距離衰減), self-similarity (自我相似性), and scale (尺度)
 - Spatial sampling (空間取樣)
- Geographic Presentation
 - Presentation: Discrete objects (離散物件) and continuous fields (連續場)
 - Digital presentation: raster data (網格資料) and vector data (向量資料)
 - Generalization (概括化)

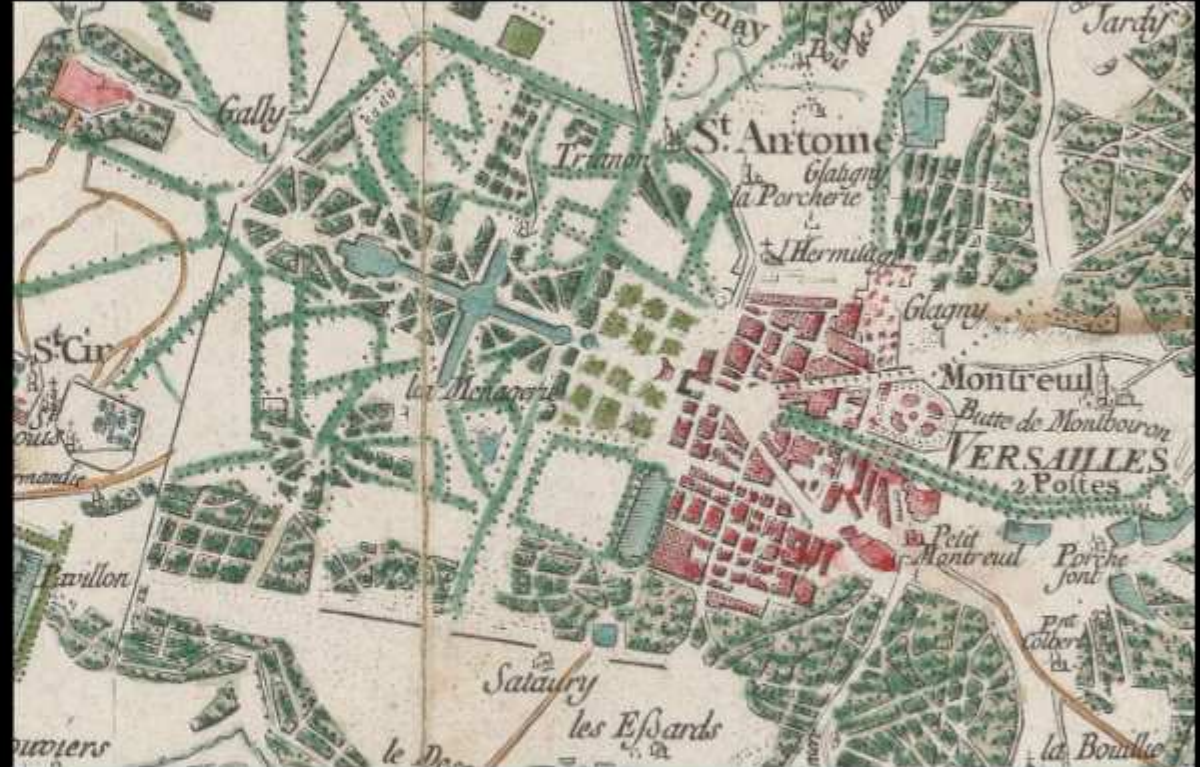
Categories of Maps

- By scale
 - Small-scale maps (regional)
 - Large-scale maps (local)
- By function
 - General reference maps
 - Thematic maps
 - Charts (navigation)
- By subject matter

18th Century: Survey Maps



Carte de Cassini, Cassini Map or Academy's Map



the pioneering topographic and geometric survey of the entire Kingdom of France, created over four generations of the Cassini family from 1750 to 1815 at a uniform scale of 1:86,400 and comprising 182 engraved sheets that could be assembled into a massive wall map approximately 39 feet high by 38 feet wide.

General Reference Map (普通地圖)

- General reference maps are to show the locations of a variety of different features, such as water bodies, coastlines, and roads.
- It is designed for orientation and context, not a single theme.
- display information **qualitatively** by using **symbols** to communicate distributions on the cultural landscape
- General reference maps are the spatial backbone of governance—supporting control, planning, identity, and power.
- 台灣常用普通地圖
 - 台灣通用電子地圖
 - 國土測繪圖資服務雲
 - 地理資訊整合圖台
 - 經建版地形圖
 - 相片基本圖

中華民國陸軍部
第一萬五千七百零一號

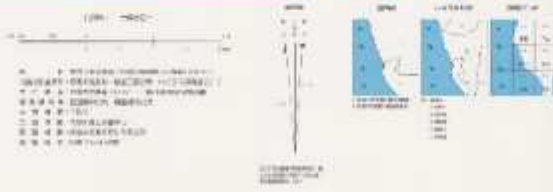
高雄市

比例尺 1:50,000



圖例

圖例	說明
1. 國界	2. 省界
3. 縣界	4. 鄉界
5. 鎮界	6. 村界
7. 主要道路	8. 次要道路
9. 鐵路	10. 水運線
11. 河流	12. 湖泊
13. 海岸線	14. 港口
15. 機場	16. 火車站
17. 公園	18. 學校
19. 工廠	20. 倉庫
21. 墳場	22. 廟宇
23. 其他	



中華民國
臺灣地區行政區劃圖

潮州

第三版 圖號 5722-111-604



圖例

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19. 工廠	20. 倉庫
21. 墳場	22. 廟宇
23. 其他	

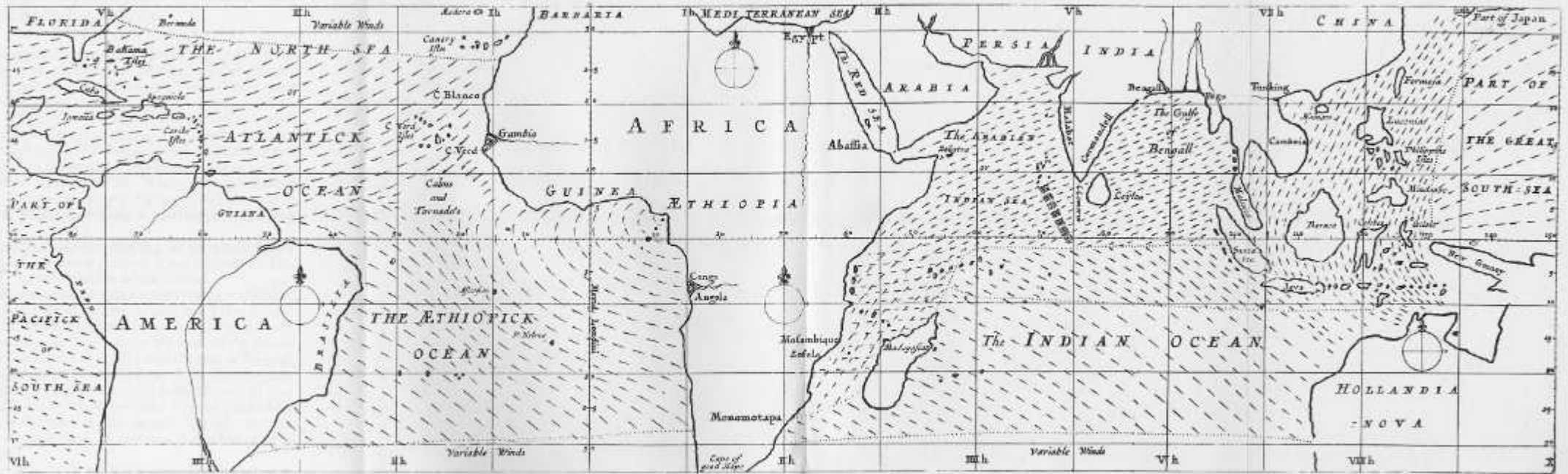




例

公路及編號：		高壓線		省轄市、縣政府、直轄市區公所	◎
國道(高速公路)		房屋區：獨立房屋		縣轄市、鄉、鎮、省轄市區公所	⊙
省道		鐵路橋		警察機關	☒
縣道		公路橋		大學：中學：小學	因 由 因
鄉道		小橋		醫院：衛生所	因 因
小徑		可通行之水壩		郵局：亭	因 因
鐵路：		不能通行之水壩		墓地：祠堂	⊥ 示
一般鐵路		河流：時令河：湖		寺廟：教堂	卐 十
輕便鐵路		沙地		塔：水塔	☒ ⑨
捷運路網		崩土		發電廠：變電所	☼ ②
高速鐵路		堤防：可通行之堤防		水廠：加油站	♀ ①
機車、架空索道		水池：養殖池：鹽田		工廠：礦場	⊕ ×
公路隧道		沼澤：溼地		電信局：電臺	☎ ③
鐵路隧道		竹林：綠地		氣象臺：電信塔	⊥ ④
漁港		茶林：闊葉林		衛星控制點：衛星定位追蹤點	△ ⑤
直轄市、省界		果園：樹林		水準點：標高點	□ ×146
省轄市、縣、直轄市區界		旱田：水田			
縣轄市、鄉、鎮、省轄市區界		直轄市：省政府			

17-18th Century: Thematic Maps

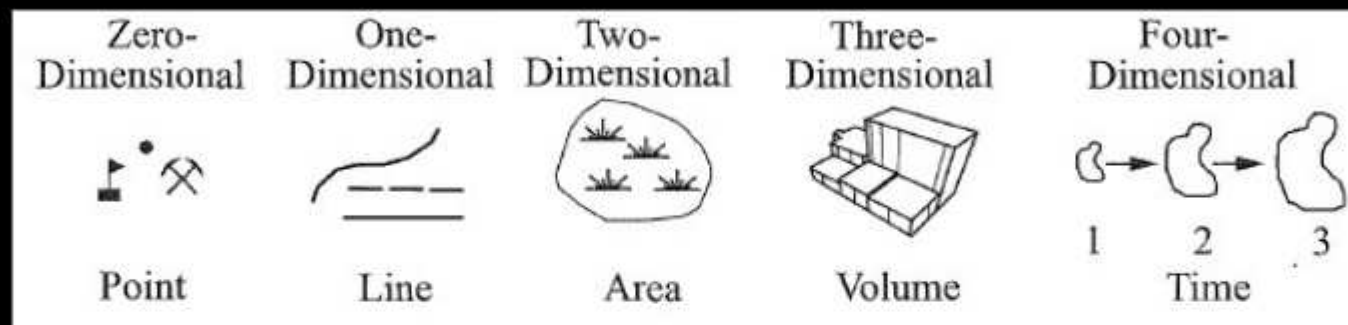


Edmond Halley made a world map of trade winds and monsoons (first meteorologic map) in 1686.

Thematic Map (主題地圖)

- Thematic maps or special purpose maps concentrates on the distribution of a single attribute or the relationship among several.
- Thematic maps involve the display of **quantitative** data that are easily incorporated into a GIS or mapping software so that we can communicate the spatial variability of that data.
- The data are presented in a row and column format in which the geographic location, whether it be for a specific location that can occur at a point, along a line, or within a polygon, is normally represented in rows and the many attributes collected for that location are presented in the columns.
- We investigate the **statistical characteristics** of the data and how they assist us in the selection of which thematic map to use.

Types of Spatial Objects



- Geographic objects are classified according to their topological dimension, which provide a measure of the way they fill space.
- A point has neither length nor breadth nor depth, and hence it is said to be of dimension 0.
- Lines have length, but not breadth or depth and hence are of dimension 1.
- Area objects have the two dimensions of length and breadth, but not depth.
- Volume objects have length, breadth, and depth and hence are of dimension 3.
- Time is often considered to be the fourth dimension of spatial objects.

Types of Thematic Map

- Point maps (點位地圖)
- Line maps (線狀地圖)
- Area maps (面狀地圖)
- Cartograms (示意地圖)
- Chartmaps (圖表地圖)
- 3D maps (三維地圖)

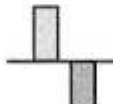
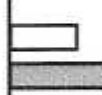
Types of Thematic Map

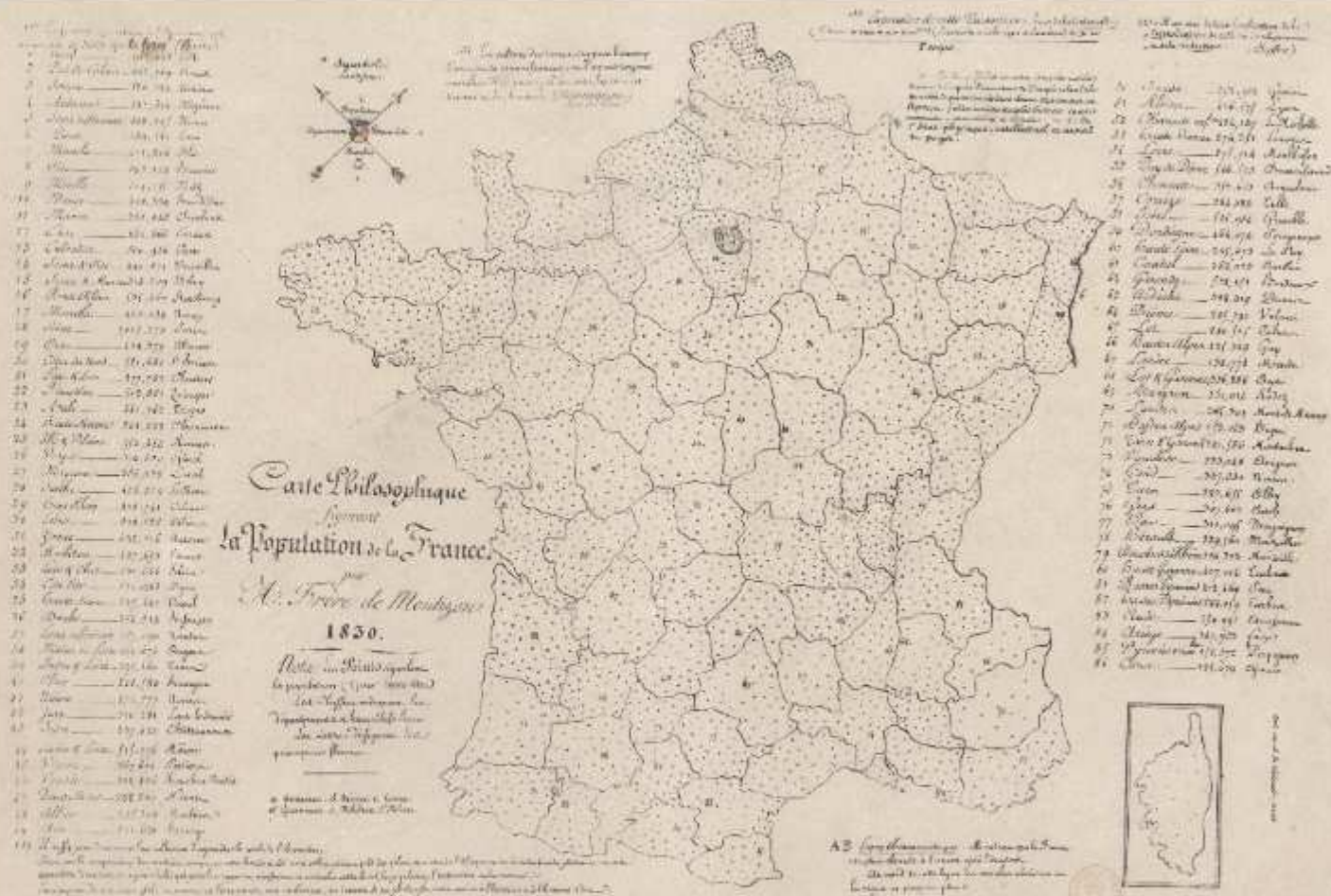
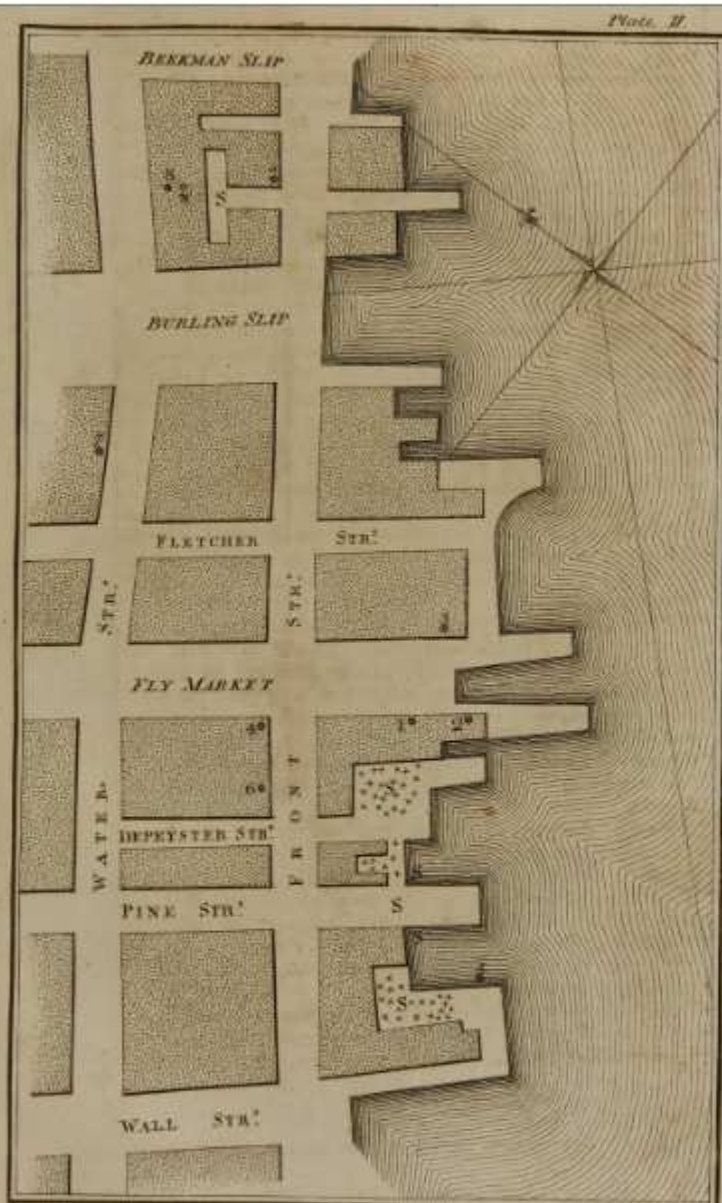
- Point maps (點位地圖)
- Line maps (線狀地圖)
- Area maps (面狀地圖)
- Cartograms (示意地圖)
- Chartmaps (圖表地圖)
- 3D maps (三維地圖)

Point Maps

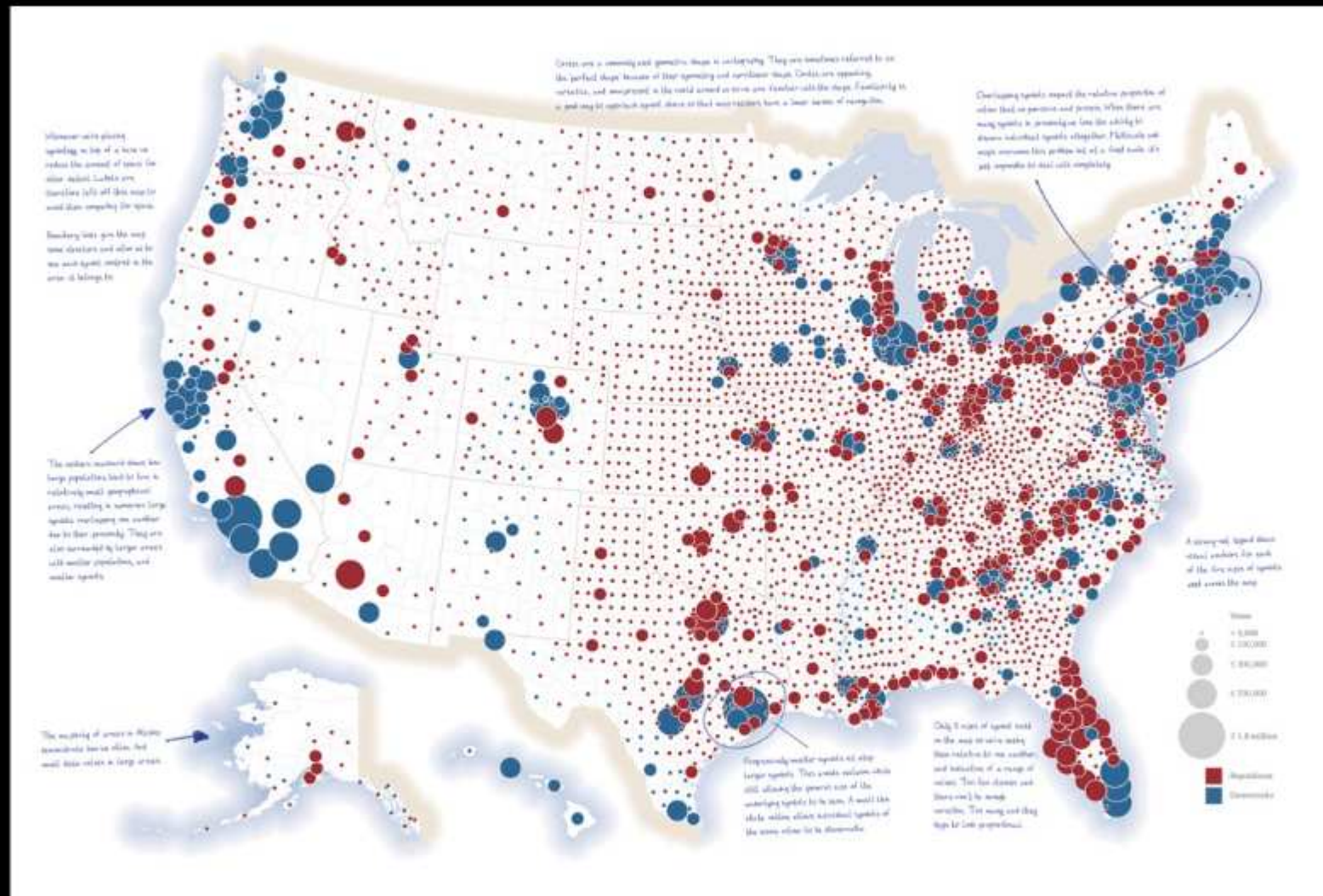
- Point-based symbol maps can work directly with the raw count data.
- The symbols additionally encode the number of counts by varying their size.
- A graduated symbol map can be considered the inverse of the classified choropleth map (面量圖).
- You can use transparency or cut-outs in the symbol design, but in a map of so many symbols, it is probably impossible to make a map devoid of overlaps.

Point Symbols

Qualitative By Location and Nature		Quantitative By Location and Amount				
Point Symbols						
	Ship Wreck					Presence
	School					
	Church					Length
	Building					
	Mine					Area
	Bench Mark					
	Airport					
	Towns and Cities					Volume

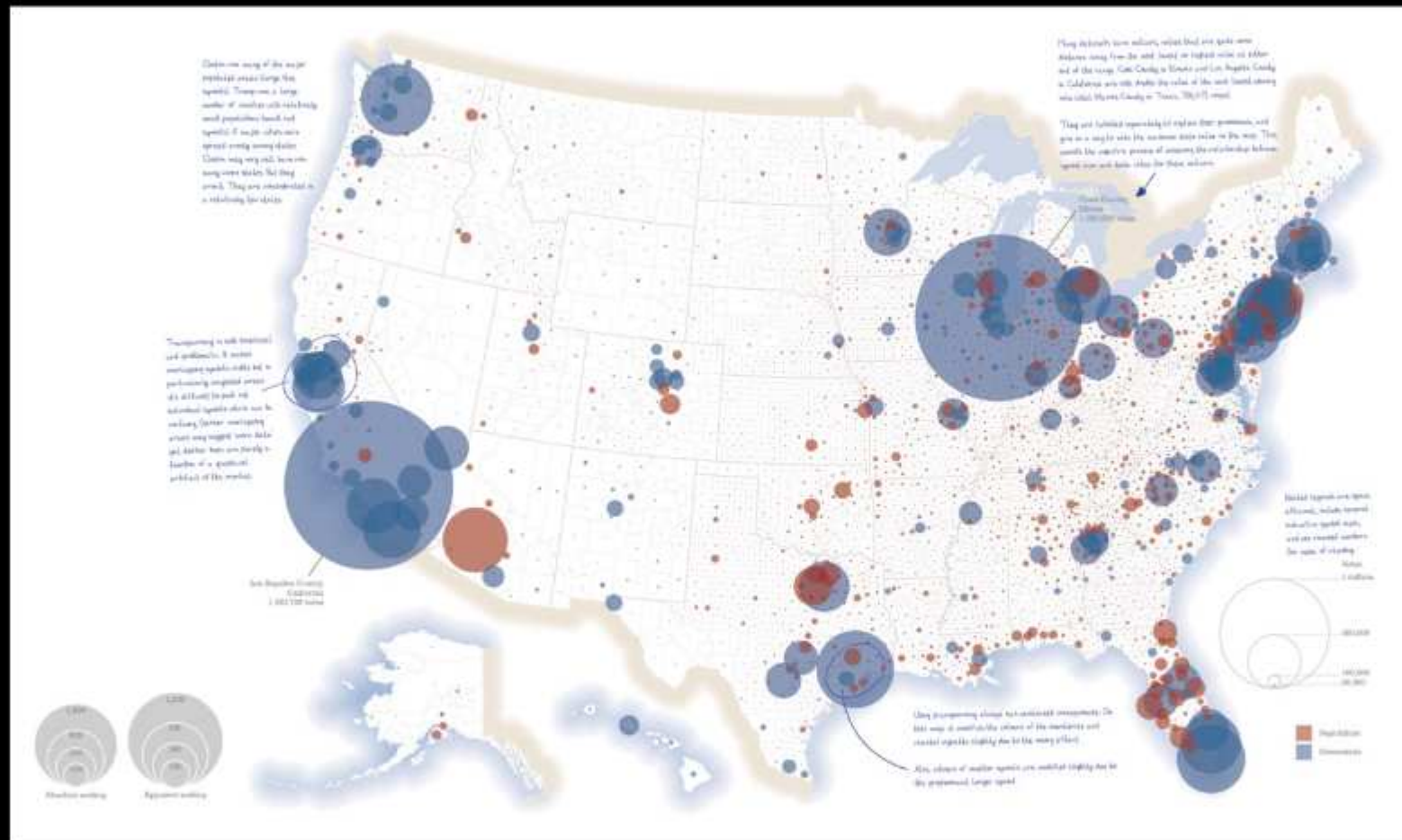


Geometric shapes graduated in size



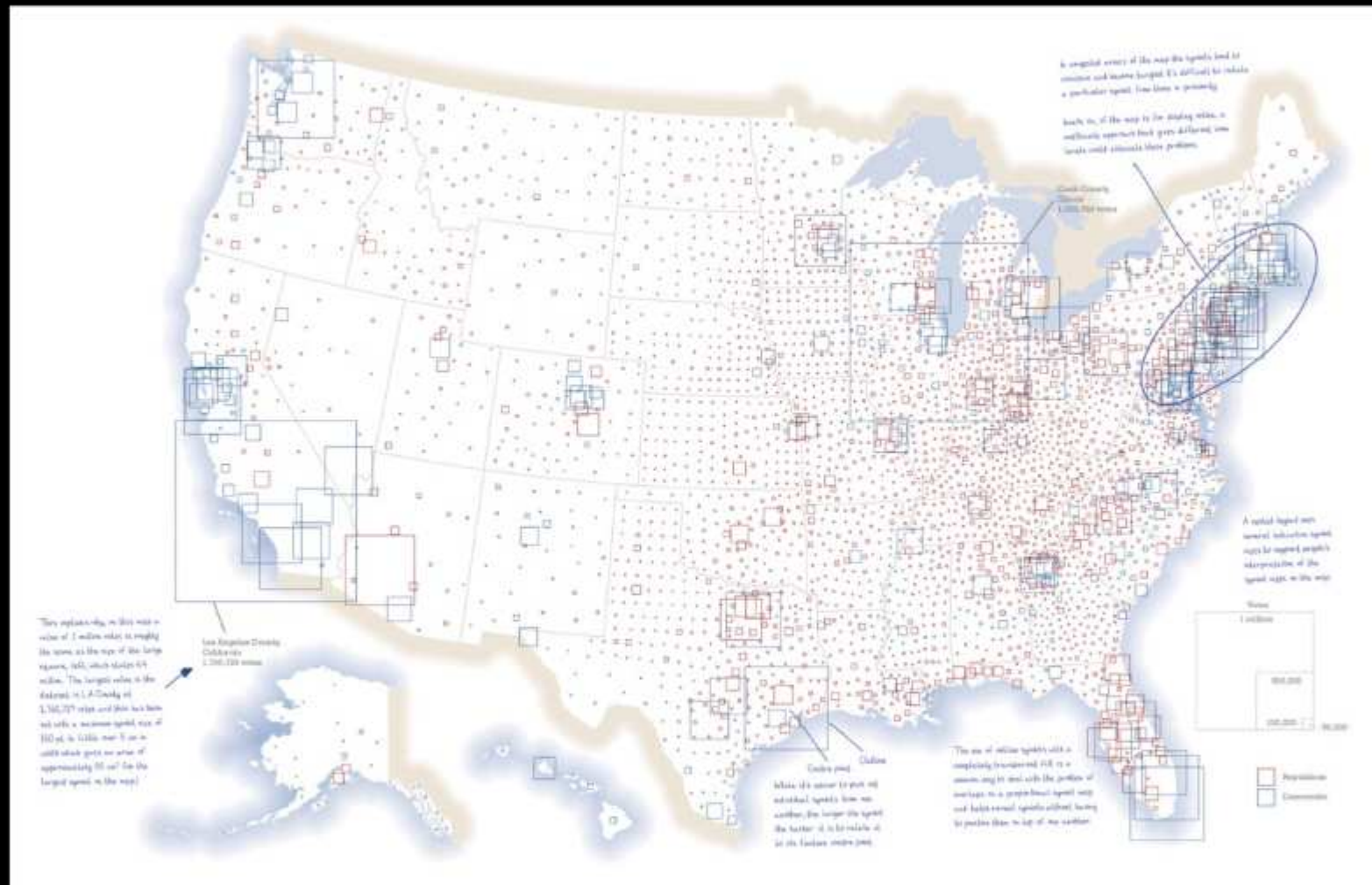
Field (2022)

Proportional symbols emphasize the data range



Field (2022)

Proportional symbol map using squares



Field (2022)

Minimum detail with isotype symbology

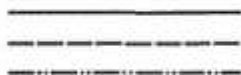
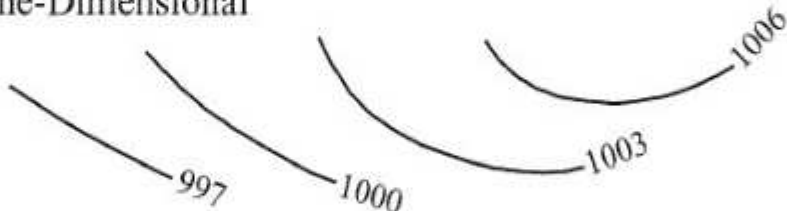


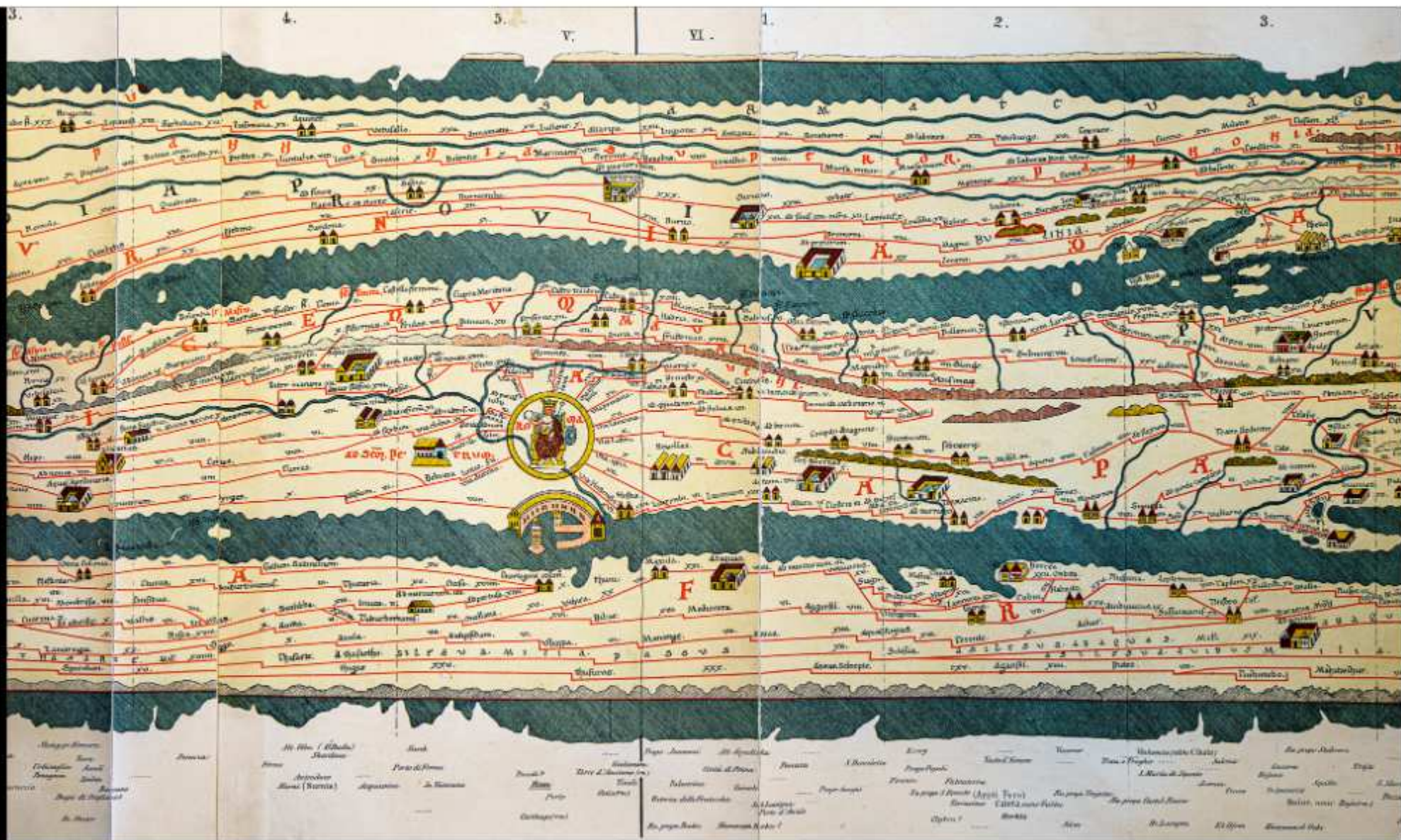
Field (2022)

Line Maps

- Lines are used as a graphical way of encoding the boundary between neighboring place.
- Line-based thematic maps are a familiar sight for representing continuous field data as isopleth maps (等值線圖).
- Isopleth maps are used to visualize phenomena that vary continuously over space. An isoline (等值線) connects points with equal attribute, such as contour lines (等高線), isohyets (等雨量線), isochrones (等時線), isodapanes (等運輸成本線).
- Lines with arrows can show moving directions, and the widths of the lines can represent the magnitude (the value of the attributes).

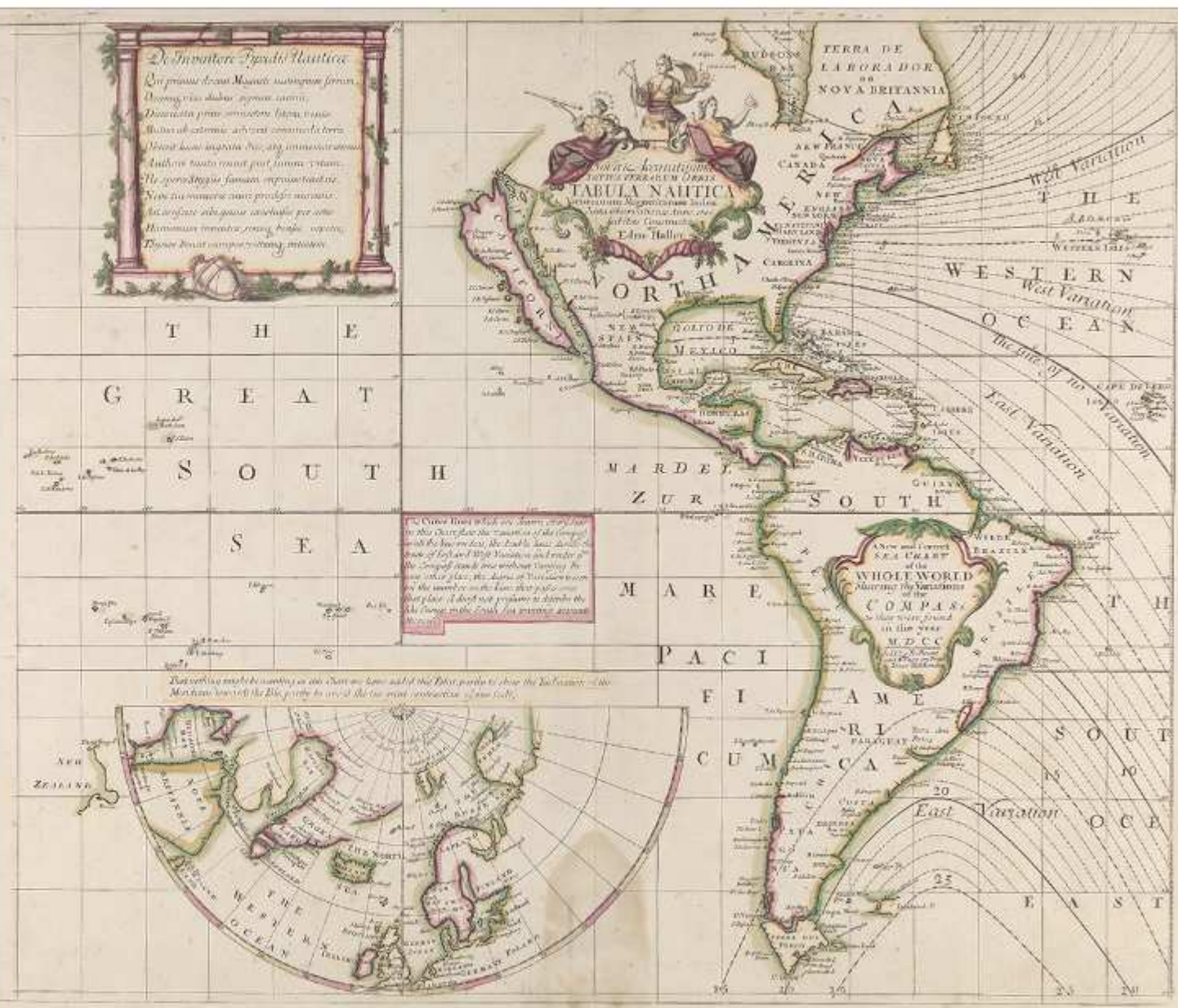
Line Symbols

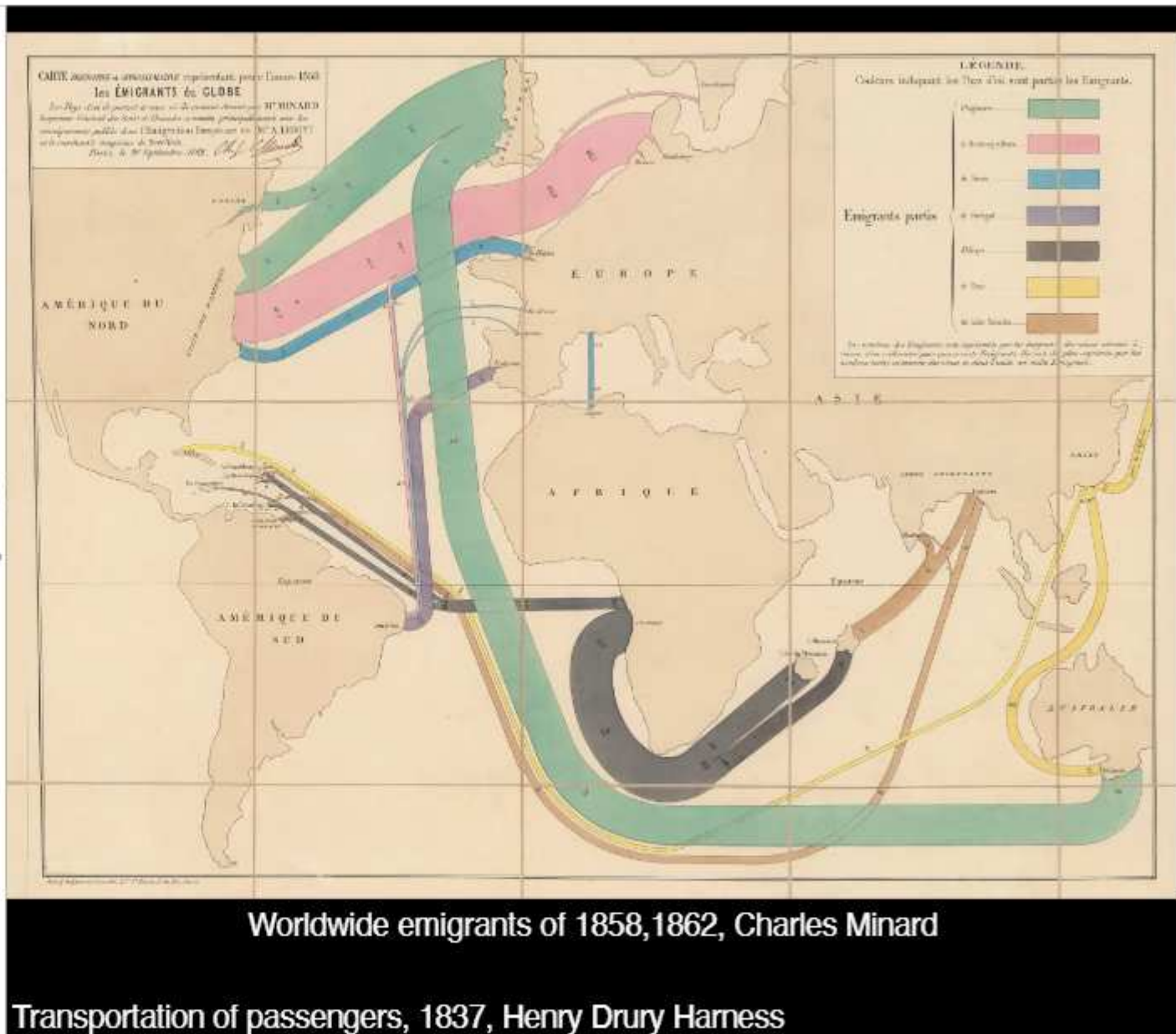
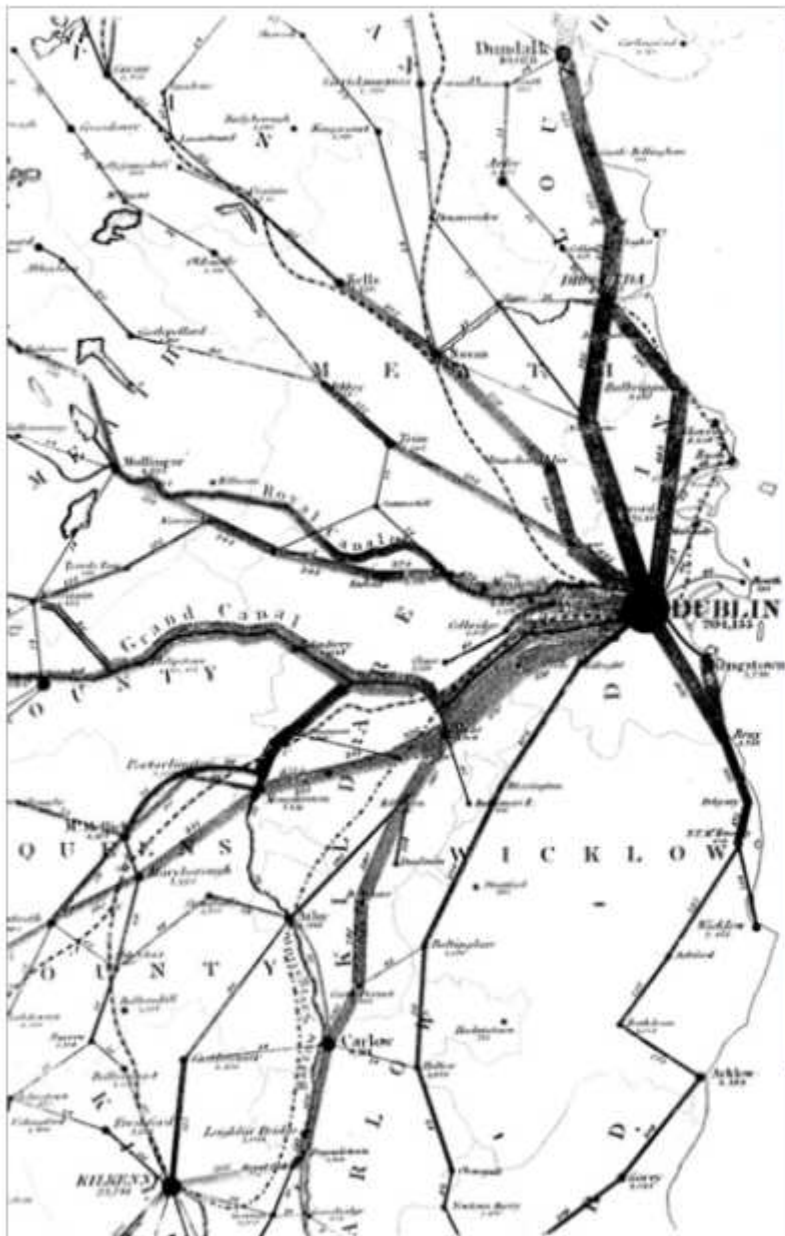
Qualitative By Location and Nature		Quantitative By Location and Amount	
Line Symbols			
 Political Boundary  Parallels and Meridians	One-Dimensional		Relative Position
 Railroad  River  Road	Two-Dimensional		Width

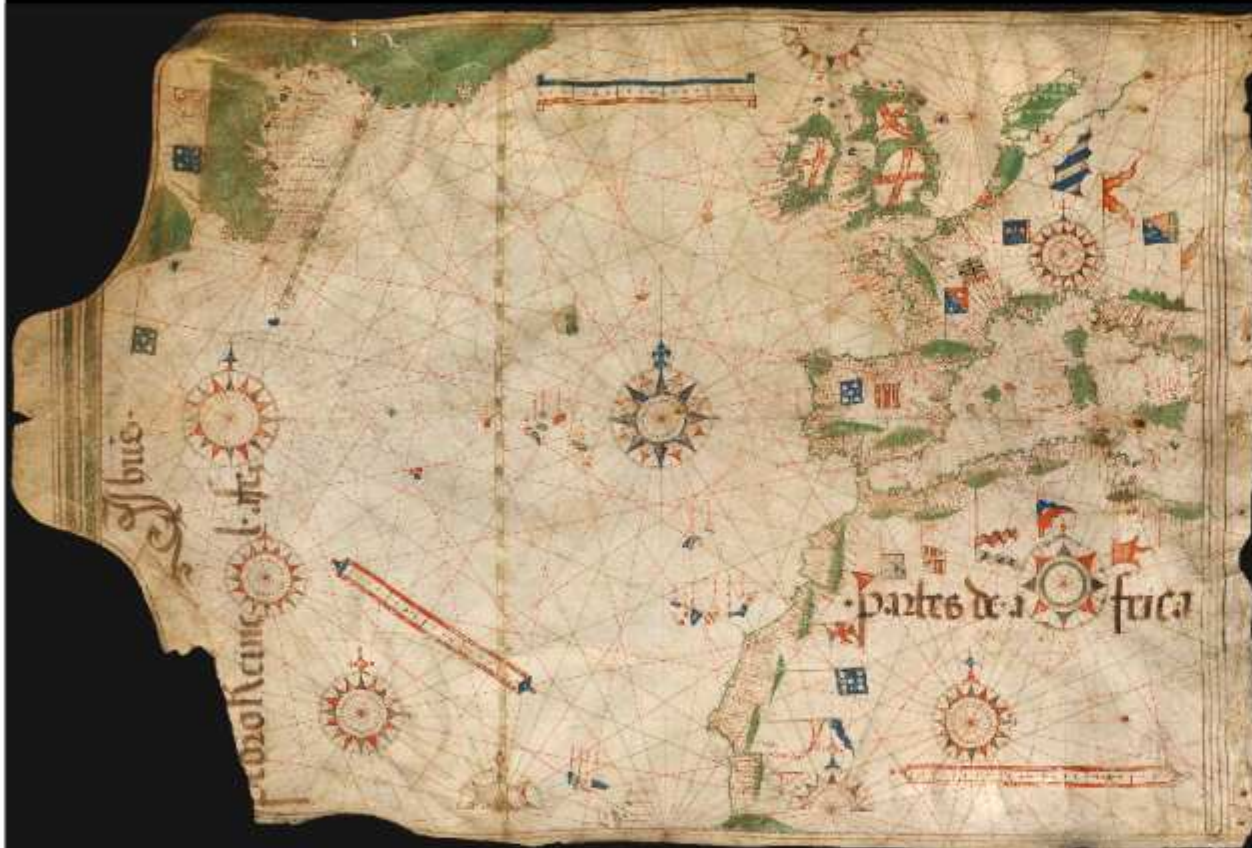


Peutinger Map, 12th Century

New and Correct Chart Showing
the Variations of the Compass,
1702, Edmond Halley







In 1504, the arrow first was used by Portuguese cartographer Pedro Reinel to indicate north on a compass rose.



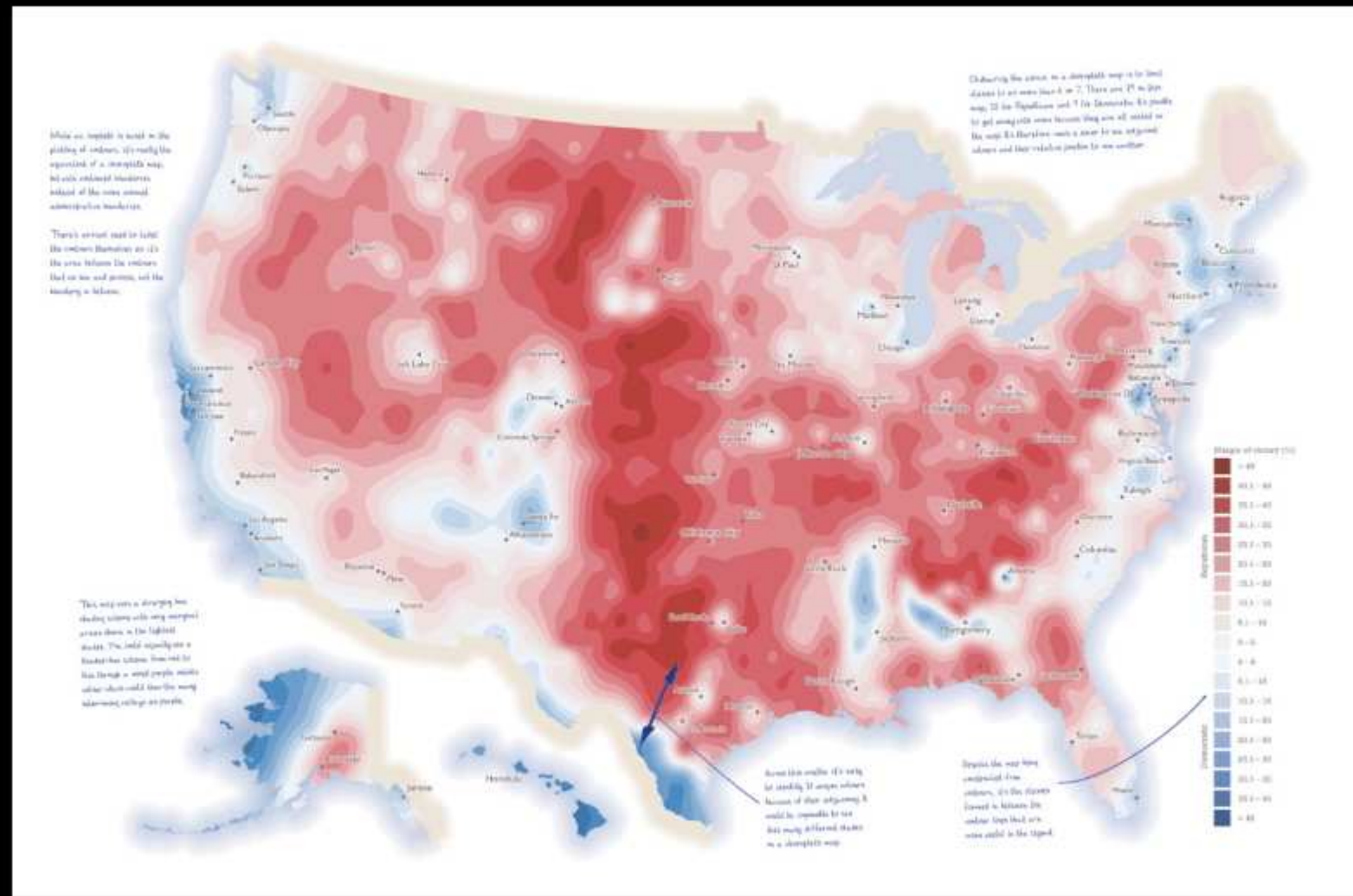
Wars of the Roses, 1903, Emil Reich

Isopleth map



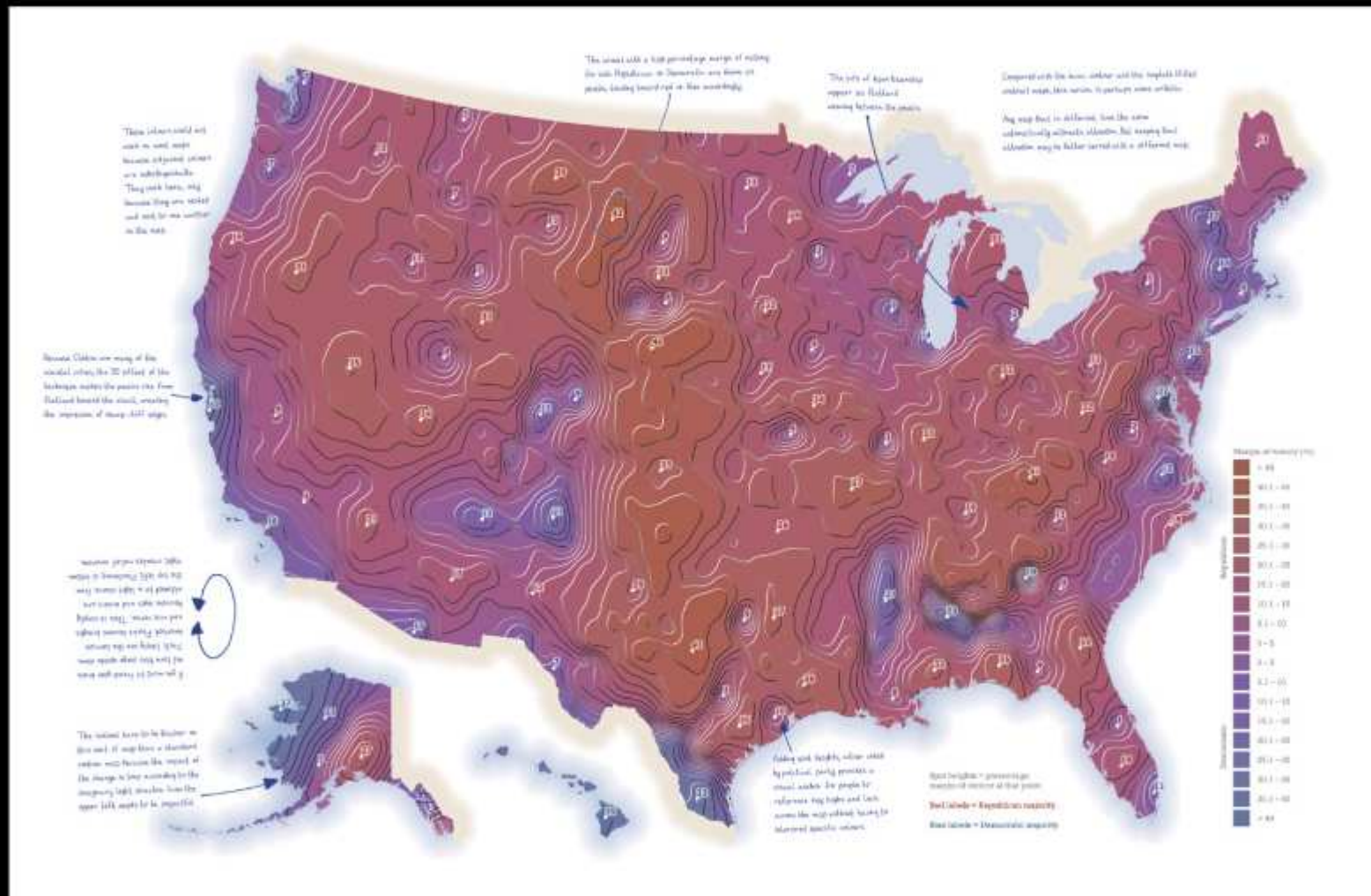
Field (2022)

Isopleth (filled isolines)



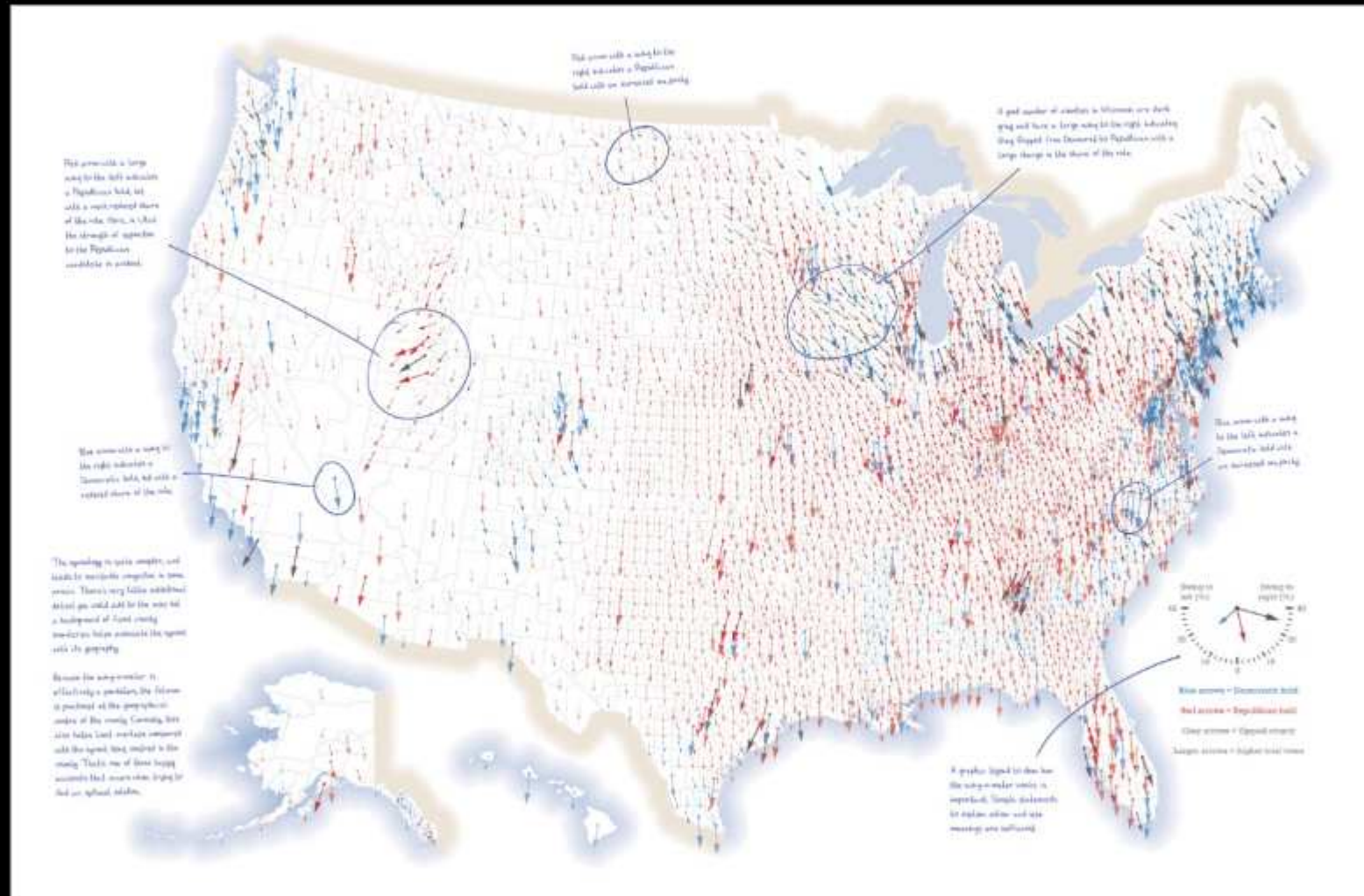
Field (2022)

Illuminated isolines



Field (2022)

Directional symbology



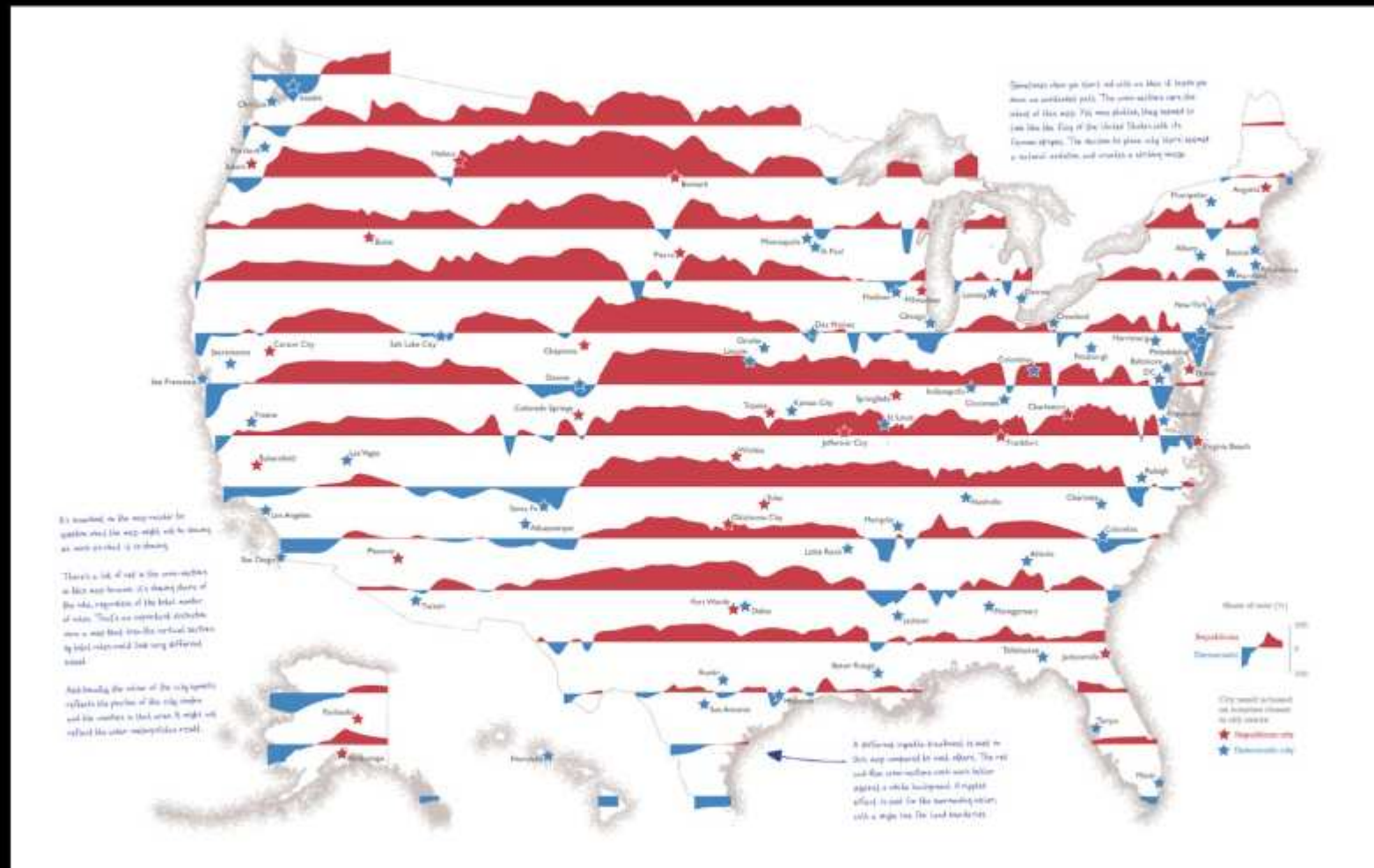
Field (2022)

Trends over time using directional arrows



Field (2022)

Exposing cross-sections across the map



Field (2022)

